



ME 300: Engineering Thermodynamics

First Statement of the Second Law

Second Law: First Statement (1A)

→ Although work can be completely converted to heat, heat cannot be completely and continuously converted to work.



{ heat $\rightarrow \Delta T$
T $\rightarrow$ mean velocity of molecules

work = force $\times$ distance

$\rightarrow$ directional

Second Law: First Statement (1B)

Kelvin-Planck: It is impossible to construct a cyclically-operating device for which the sole effect is the exchange of heat from a single reservoir and creation of the equivalent amount of work.

→ cyclically operating = continuously

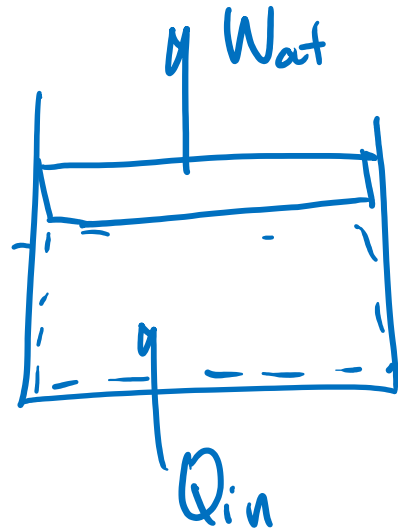
→ reservoir = source of heat with a constant temp.
sink



Does not apply to processes

Statement 1 applies to cycles, not processes.

Example: isothermal expansion of ideal gas

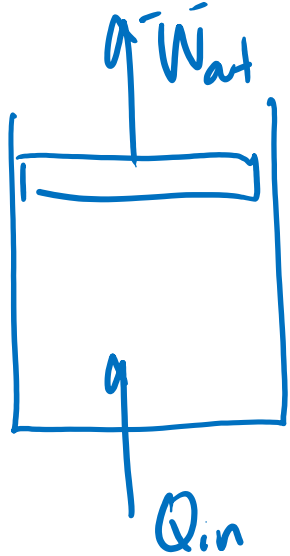




What is the change in internal energy for this process?

- A. Greater than zero
- B. Less than zero
- C. Equal to zero

Does not apply to processes



isothermal $\rightarrow \Delta T = 0$

ideal gas $\rightarrow \Delta u = c_v \Delta T = 0$

$\Rightarrow \Delta KE = \Delta PE = 0$

$\Rightarrow \Delta \bar{U} = Q - \bar{W} = 0 \Rightarrow \text{process: } Q = \bar{W}$

~~*~~ does not violate 2nd law



What is the change in internal energy for this process if the fluid starts as a liquid and ends as a vapor?

- A. Greater than zero
- B. Less than zero
- C. Equal to zero

$$\Delta u > 0, \Delta u = u_g - u_f$$